

Charles Hu

McLean, VA | (408) 636-8544 | Charley2024@gmail.com | linkedin.com/in/charles-hu-0639a6220

EDUCATION

Carnegie Mellon University — B.S., Electrical and Computer Engineering Graduating May 2028

WORK EXPERIENCE

Intern, Pylee AI June 2024 – August 2024

- Designed and implemented MCP servers to support scalable AI system architectures.
- Integrated and evaluated LLM interfaces through structured testing workflows to improve system reliability.

Intern, Fida Labs April 2023 – May 2023

- Processed and analyzed patient samples for HIV testing using standardized laboratory protocols.
- Maintained strict documentation and biosafety compliance to ensure data accuracy and confidentiality.
- Supported day-to-day laboratory operations, gaining practical experience in diagnostic testing workflows.

RESEARCH

Research Assistant, Zoe-2 Planetary Rover — Planetary Robotics Lab January 2025 – Present

- Designed and implemented the electrical architecture of a planetary rover platform, integrating motors, encoders, and CAN bus communication systems.
- Conducted field experiments comparing passive vs. skid steering rover architectures, analyzing drivetrain performance and efficiency.
- Developed a real-time robotics telemetry and debugging pipeline using motion capture and Foxglove visualization.
- Supported robot subsystem integration and validation for mobility platform testing.

Research Assistant, Carnegie Mellon CREATE Lab December 2025 – Present

- Developed a structured human-AI interaction curriculum for K-5 students, increasing accessibility and engagement in AI literacy education.
- Conducted research on AI-adolescent interaction patterns, synthesizing findings to inform educational and developmental strategies.
- Investigated technology fluency trends in modern society, contributing to research-driven frameworks for AI education.

Research Assistant, Carnegie Mellon EDL Lab January 2026 – Present

- Conducting experimental research on hydrogen fuel cell systems, analyzing electrochemical performance, efficiency, and degradation mechanisms.

LEADERSHIP & ACTIVITIES

NCAA Varsity Athlete, Carnegie Mellon Swim Team September 2024 – Present

- Balanced 20+ hours/week of NCAA training with a rigorous engineering course load while maintaining strong academic performance.
- Contributed to team success in high-pressure competitions through discipline, teamwork, and resilience.

Co-President, AYLUS September 2022 – May 2024

- Served as President of a volunteer organization after five years of active involvement.
- Coordinated and led community service initiatives to maximize outreach and impact.
- Managed and organized volunteer teams, delegating responsibilities and ensuring effective execution of events.

SKILLS

Electrical Engineering: Embedded Systems, Microcontrollers, CAN Bus, FPGA Development, Circuit Design, PCB Design & Layout, SPI, I2C, UART

Hardware & Prototyping: Soldering, CNC Machining, Machine Tool Fluency, Power Tool Fluency, Power Systems, Signal Analysis, Data Analysis, System Modeling, Robotics System Integration, Debugging Hardware, Motor Control, Sensor Integration

Programming: Python, C, C++, ROS2, VHDL, Verilog, SystemVerilog

Tools: Linux, Foxglove, SolidWorks, MATLAB, AutoCAD, Ansys Fluent, VS Code, Ubuntu, Vim, Vivado